

# Empirical Proof Record: INCONCLUSIVE

2026-05-22T19:23:53.588242+00:00

## Empirical Proof Record — **INCONCLUSIVE**

**Proof ID:** 6209cb915c832544fec4425b089877e6...

**Created:** 2026-05-22T19:23:53.588242+00:00

### 1. Claim

On real market returns, Kimera's memory tracks the order-dependent maximum drawdown that set-based retrieval is structurally blind to:  $\text{drawdown\_tracking\_rho} = \text{Spearman}(\text{kimera\_state\_divergence}(\text{true}, \text{shuffle}), |\text{maxdrawdown}(\text{true}) - \text{maxdrawdown}(\text{shuffle})|) > 0$ , while the RAG aggregate representation divergence between orderings is  $\equiv 0$ .

- **Operationalisation:** For 6 real-market return windows (len 6), run the true order and 3 shuffles through Kimera entity · batch (each a fresh trajectory). Per (window, shuffle):  $\Delta\text{MDD} = |\text{MDD}(\text{true}) - \text{MDD}(\text{shuffle})|$  (order-dependent ground truth);  $\text{kimera\_state\_divergence}$  = relative distance of the final manifold state (coupling/order/mass).  $\text{drawdown\_tracking\_rho}$  = Spearman over all pairs. RAG contrast: mean-pooled TF-IDF representation divergence between orderings ( $\equiv 0$ , same multiset). **VALIDATED** iff  $\text{rho} > 0$  with  $p < 0.05$ .
- **Threshold:**  $\text{drawdown\_tracking\_rho} > 0.0$   $\text{spearman\_rho}$
- **H0:** H0:  $\text{drawdown\_tracking\_rho} \leq 0$  — Kimera's representation does not track order-dependent drawdown; its text encoding misses the numeric path (like a set-based retriever)
- **H1:** H1:  $\text{drawdown\_tracking\_rho} > 0$  — Kimera's manifold deformation tracks the real order-dependent drawdown that set-based retrieval cannot represent

## 2. Verdict

**INCONCLUSIVE** — observed 0.18257506994942782  
drawdown\_tracking\_rho +0.1826 (p=0.23418904354012834) over 18 real-market  
window×shuffle pairs: Kimera state-divergence vs  $|\Delta\text{MDD}|$  (Spearman). RAG  
aggregate representation divergence 0.000000 (order-invariant — carries zero  
drawdown order-information). Kimera mean state-divergence 0.0181, prime-  
address order-divergence 0.8904; series FRED SP500 (real market); INCON-  
CLUSIVE — rho>0 not statistically significant

## 3. Evidence

| Pillar                               | Statistic             | Value               | 95% CI |
|--------------------------------------|-----------------------|---------------------|--------|
| 0.comparison.finance_path_dependence | drawdown_tracking_rho | 0.18257506994942782 | —      |

## 4. Pre-registration

- Registered at: 2026-05-22T19:23:53.227477+00:00
- Config hash: c76cceb25b8bc01252f5dd123d1fcede...
- Data hash: c97935afa07672f2367f5be6fcddc25e...

## 5. Signature

8830caef21eca1ab6f7b27aee656a84e8b737d2d463c551f0a6847a4c71d4ed9